1️⃣ Sales & Demand Forecasting

📌 Olist Brazilian E-Commerce Dataset

Purpose: Contains customer orders, products, delivery dates, and customer information. Useful for generating realistic customer demand.

🔗 https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

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📌 DataCo Smart Supply Chain Dataset

Purpose: Contains sales orders, products, shipping, customers, and logistics information. Useful for Sales, Warehouse, and Purchase Agents.

🔗 https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

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2️⃣ Inventory & Warehouse Management

📌 Inventory Dataset

Purpose: Contains stock levels, reorder quantities, lead times, warehouse inventory, and scheduler information.

🔗 https://www.kaggle.com/datasets/veeralakrishna/inventory

3️⃣ Procurement & Supplier Management

📌 Procurement KPI Analysis Dataset

Purpose: Contains supplier performance, procurement costs, delivery performance, and purchasing KPIs.

🔗 https://www.kaggle.com/datasets/shahriarkabir/procurement-kpi-analysis-dataset

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4️⃣ Production Planning & Control (PPC)

📌 Brandimarte Flexible Job Shop Benchmark

Purpose: Contains production jobs, operation sequences, and multiple machine choices for each operation. Ideal for flexible production planning.

🔗 https://scheduleopt.github.io/benchmarks/fjsplib

📌 **Taillard Job Shop Benchmark**

Purpose: Large-scale production scheduling benchmark used to evaluate scheduling algorithms.

🔗 https://github.com/tamy0612/JSPLIB

📌 **Fisher-Thompson (FT) Instances**

Purpose: Small benchmark scheduling problems used to validate scheduling algorithms.

🔗 https://github.com/tamy0612/JSPLIB

📌 **Lawrence (LA01–LA40) Instances**

Purpose: Standard production scheduling benchmark widely used in research.

🔗 https://github.com/tamy0612/JSPLIB

📌 **Adams-Balas-Zawack (ABZ) Instances**

Purpose: Classical job shop scheduling benchmark used for baseline algorithm comparison.

🔗 https://github.com/tamy0612/JSPLIB

5️⃣ **Flexible Job Shop Scheduling (FJSP)**

📌 **Hurink Flexible Job Shop Instances**

Purpose: Flexible scheduling problems where one operation can be processed by multiple machines.

🔗 https://scheduleopt.github.io/benchmarks/fjsplib

📌 **Kacem Flexible Job Shop Instances**

Purpose: Highly flexible scheduling benchmark used for testing machine negotiation and load balancing.

🔗 https://scheduleopt.github.io/benchmarks/fjsplib

📌 **Fattahi Flexible Job Shop Instances**

Purpose: Medium-sized flexible scheduling problems for validating optimization algorithms.

🔗 https://scheduleopt.github.io/benchmarks/fjsplib

6️⃣ **Dynamic Scheduling & Factory Logistics**

📌 **LOS (Large Scale Online Scheduling) Dataset**

Purpose: Contains setup times, transportation times, and dynamic scheduling information for realistic factory execution.

🔗 https://www.researchgate.net/publication/390101555

📌 **Multi-Objective Flexible Job Shop Cell Scheduling Dataset**

Purpose: Contains workstation-based scheduling where machines are grouped into manufacturing cells.

🔗 https://doi.org/10.1016/j.dib.2024.109946

📌 **Dauzère-Pérès & Paulli (DPP)**

Purpose: Multi-processor scheduling benchmark for testing production bottlenecks.

🔗 Available through academic benchmark repositories

📌 **Barnes & Chambers (BCData)**

Purpose: Classical scheduling benchmark used to compare modern scheduling techniques.

🔗 Available through academic scheduling repositories

7️⃣ **Machine & Predictive Maintenance**

📌 **AI4I 2020 Predictive Maintenance Dataset**

Purpose: Contains machine sensor readings and machine failure information for predictive maintenance.

🔗 https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset

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📌 **NASA C-MAPSS**

Purpose: Aircraft engine degradation dataset used for advanced predictive maintenance research.

🔗 https://data.nasa.gov/

🎱 **Quality Inspection & Defect Detection**

📌 **UCI SECOM Dataset**

Purpose: Manufacturing sensor data with pass/fail quality labels for quality prediction.

🔗 http://archive.ics.uci.edu/ml/datasets/SECOM

📌 **MVTec AD**

Purpose: Industrial product images with normal and defective samples for visual defect detection.

🔗 https://www.mvtec.com/research-teaching/datasets/mvtec-ad

📌 **MVTec 3D-AD**

Purpose: 3D industrial object scans with defects for advanced quality inspection.

🔗 https://www.mvtec.com/research-teaching/datasets/mvtec-3d-ad

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📌 **MVTec LOCO AD**

Purpose: Industrial images containing logical assembly errors such as missing or misplaced components.

🔗 https://www.mvtec.com/research-teaching/datasets

9️⃣ Resource & Workforce Planning

📌 PSPLIB (Project Scheduling Library)

Purpose: Resource-constrained scheduling benchmark used for planning labor, machines, and other limited resources.

🔗 http://www.om-db.wi.tum.de/psplib/

🔟 Large Industrial Manufacturing Dataset

📌 Bosch Production Line Performance Dataset

Purpose: Large-scale real manufacturing dataset containing production line sensor readings and quality information.

🔗 https://www.kaggle.com/c/bosch-production-line-performance